

Problem statement

Requirements

Actors

Use Stories

Use Cases

Use case diagram

Project SDLD from my perspective

1. Problem Statement
2. Requirements
3. Use Cases
4. Actors
5. Use Cases
6. User Stories
7. Use case Diagrams
8. Class diagram
9. ERD
10. UI Design sketch
11. Architecture Sketch
12. Tech Stack Choice

Architecture Design

Tier Architecture (Database – Backend – Frontend)

Database:

Database analysis

1. core entities
2. core properties
3. relationships
4. constraints
5. processes or use cases
6. time and history tracking
7. OLAP and O
8. data ownership
9. Time & History Tracking
10. privacy and GDPR
11. Access control
12. AuditLogs
13. Queries-what the database will answer

Database Implementation:

1. Create the database
2. Create the tables
3. Create the relationships
4. Create the constraints
5. Create the required procedures and functions
6. Create the views or output required
7. Create the procedures of the queries
8. Create the triggers needed

Backend

1. Install required tools and IDEs and servers
2. Create the github repo
3. create the project folder
4. clone the repo
5. build the skeleton
6. Build Infrastructure layer
7. Build Business layer
8. Build UI Layer
9. Bind the Business with UI Layer

Introduction

Purpose

Intended audience and reading suggestions

scope

references

overall description

Business Requirements

Business Goals

Business Facts

Business Events

Business Processes

Business Rules

Product Requirements

Product Features

Product Components

Product Input

Product Processes

Product Output

Functional Requirements

Non-Functionanl requirements

Users Requirements

System requirements

product features or service features

user classes and characteristics

design and implementation constraints

Regulatory or Corporate Policies

Hardware Constraints

Interfaces and Integration:

Ex sys nam e	Integeratio n type	Dat a flow dire	Data forma t	frequenc y	Auth required ?	error handlin g strateg y	dependenci es

Technology Stack Constraints:

Item	description
Approved langauges	
disallowed tools	
required frameworks	
hosting requirements	
database technology	
platform dependency	
version constraints	

Operational Constraints:

Item	demo
uptime requirement	
backup strategy	
deployment method	
maintenance windows	
monitoring tools	
support hourse	
rollback plan	

Language and Localization:

Item	desc
Supported Languages	
default language	
right to left support	
locale-specific formats	
translation approach	
regional legal requirements	

Security Requirements:

Item	desc
authentication method	
authorization strategy	
data encryption	
senseitive data handling	
secure coding standards	
logging and audit trails	
incident response plan	

Communication Protocols

Item	desc
protocol name	
purpose	
ports/ firewall settings	
encryption used	
bandwidth requirements	
latency sensitivity	

Coding and Design Standards

Item	desc
Code style guide	
naming conventions	
design patterns allowed	
code review requirements	
documentation standards	
testing requirements	
reusability and modularity	

assumptions and dependencies

System Features

System Features:

Feature Name	Priority	Description	User story

system feature name, description and priority , use case brief description

System Models

Introduction to System Models

Purpose of modeling

Benefit of modeling

Types of models included

Use Case Model

Use Case Diagram

Use case lists

Use Case Descriptions

Item	Description
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Use Case ID	
Use Case Name	
Priority	
Description	
Goal	
User Story	
Actor	
Related Use cases or processes(dependencies	
Preconditions	
Postconditions	
Normal course of events	
Alternative courses	
Exceptions	
Includes	
Special requirements	
Assumptions	
Notes and issues	

Item	Description
Use Case ID	
Use Case Name	
Priority	
Description	
Goal	
User Story	
Actor	
Related Use cases or processes(dependencies	
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Normal course of events	
Alternative courses	
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Related Use cases or processes(dependencies	
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Normal course of events	
Alternative courses	
Exceptions	
Includes	
Special requirements	
Assumptions	
Notes and issues	

Class Diagram

Activity Diagrams

Sequence Diagram

State Diagram

Data Models

ERD

EERD

Relational Model

Physical Model

Component Diagram

System Architecture Diagram

Data Requirements

Entity name							
description							
Attribute	Data type	Length	pK	FK	Is Unique	Is Null	

appendices

introduction :

overall description

system features:

data requirements:

data dictionary:

entity name4545

table for:

Attribute,Data type, not null, PK,FK,DFK,ChkK,UQK,Description

Product Name

Product Purpose

Version

Release Date

System description

scope of coverage: features

out of scope elements

intendend audience

purpose of this version

document status

description of the software being specified and its purpose, relevant benefits, objectives, and goals, related the software to the corporate goals

Origin of the product

-why this being developed?

-what problem it solve

-does it a development only

Context:

Dos this project part of a large system

-where does it used?

organizational goals:

what is the p

interaction with external systems

Product features:

Feature ID, Feature Name, Description, Primary users, Related Modules

2.3. User Classes and Characteristics